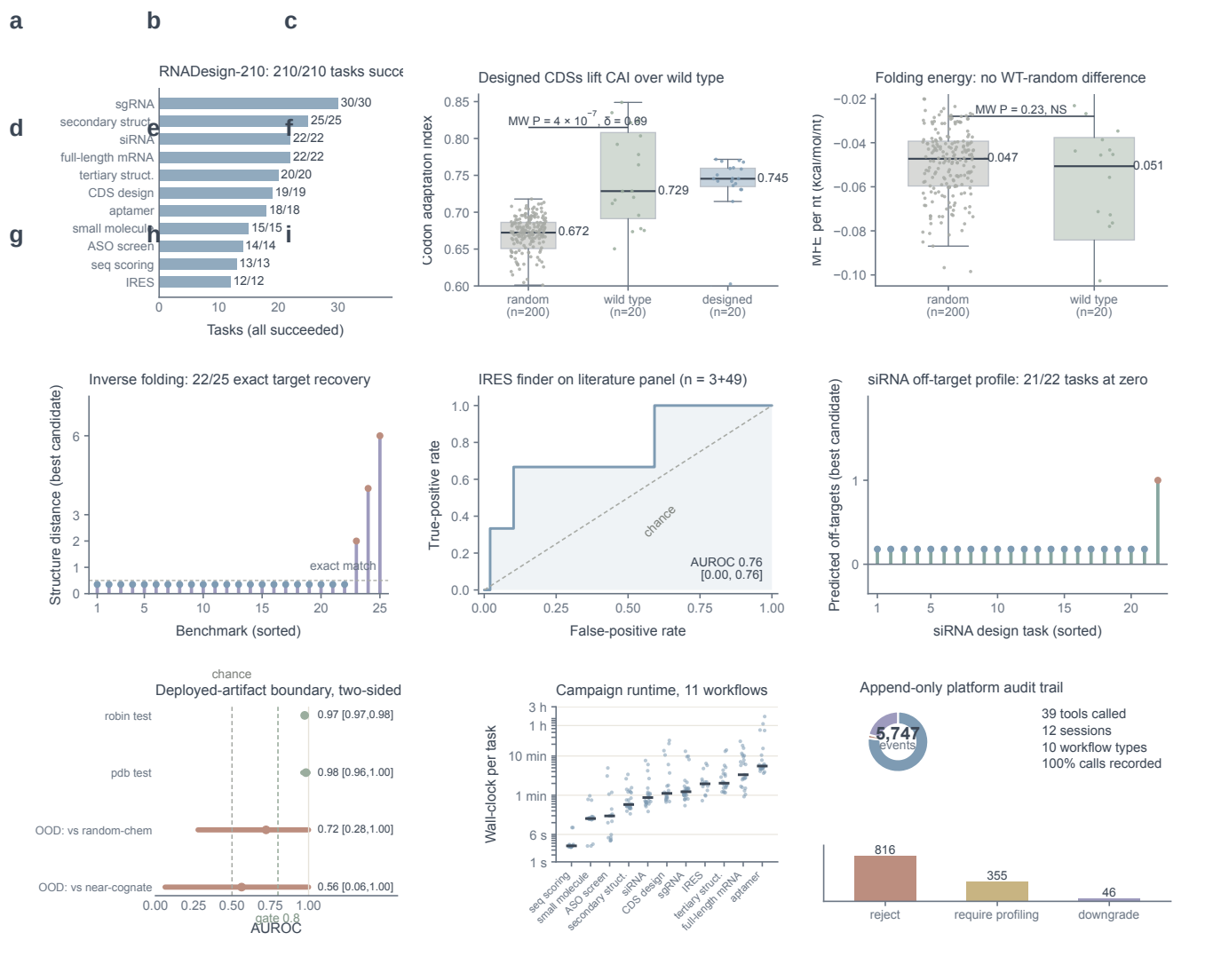

a
b
c
RNADesign-210: 210/210 tasks succe
sgRNA
secondary struct.
siRNA
full-length mRNA
tertiary struct.
CDS design
aptamer
small molecule
ASO screen
seq scoring
IRES
30/30
25/25
22/22
22/22
20/20
19/19
18/18
15/15
14/14
13/13
12/12
0
10
20
30
Tasks (all succeeded)
d
e
f
g
h
i
Designed CDSs lift CAI over wild type
MW P = 4 × 10⁻⁷ × δ = 0.89
0.672
0.729
0.745
Codon adaptation index
random (n=200)
wild type (n=20)
designed (n=20)
Folding energy: no WT-random difference
MW P = 0.23, NS
0.047
0.051
MFE per nt (kcal/mol/nt)
Inverse folding: 22/25 exact target recovery
exact match
Structure distance (best candidate)
Benchmark (sorted)
IRES finder on literature panel (n = 3+49)
chance
AUROC 0.76
[0.00, 0.76]
True-positive rate
False-positive rate
siRNA off-target profile: 21/22 tasks at zero
Predicted off-targets (best candidate)
siRNA design task (sorted)
chance
Deployed-artifact boundary, two-sided
robin test
pdb test
OOD: vs random-chem
OOD: vs near-cognate
0.97 [0.97,0.98]
0.98 [0.96,1.00]
0.72 [0.28,1.00]
0.56 [0.06,1.00]
gate 0.8
AUROC
Campaign runtime, 11 workflows
Wall-clock per task
3 h
1 h
10 min
1 min
6 s
1 s
seq scoring
small molecule
ASO screen
secondary struct.
siRNA
CDS design
sgRNA
IRES
tertiary struct.
full-length mRNA
aptamer
Append-only platform audit trail
5,747
events
39 tools called
12 sessions
10 workflow types
100% calls recorded
816
355
46
reject
require profiling
downgrade